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COURSE NAME:JAVA PROGRAMMING

COURSE CODE:CSA 0924

CO2-AT1

A plugin-based application loads objects from several vendors. Some vendors incorrectly implement equals() without a matching hashCode().

Predict the impact on HashSet and HashMap and propose testing checks. [Equality Contract].

Aim:

To demonstrate the importance of implementing both equals() and hashCode() correctly and observe their behavior in HashSet and HashMap.

Pseudocode:

START

Create two objects A and B

IF A.equals(B) is true THEN

 Check whether A.hashCode() == B.hashCode()

END IF

Add A and B to HashSet

Display HashSet size

Add A and B as keys in HashMap

Display HashMap size

STOP

Java Code:

```
import java.util.*; class Student
{
    int id;

    Student(int id) {        this.id
= id;    }

    @Override    public boolean
equals(Object obj) {        Student s =
(Student) obj;        return this.id ==
s.id;
    }

    // hashCode() is intentionally NOT overridden
} public class Main {    public static void
main(String[] args) {        Student s1 = new
Student(101);

        Student s2 = new Student(101);

        System.out.println("equals(): " + s1.equals(s2));        HashSet<Student>
set = new HashSet<>();

        set.add(s1);        set.add(s2);

        System.out.println("HashSet Size: " + set.size());

        HashMap<Student, String> map = new HashMap<>();

        map.put(s1, "First");        map.put(s2,
"Second");

        System.out.println("HashMap Size: " + map.size());
    }
}
```

Input:

Student 1 ID = 101 Student

2 ID = 101

Output:

equals(): true

HashSet Size: 2

HashMap Size: 2


Online Java Compiler

 Great ideas deserve a **great domain name**.

Main.java



```
1 import java.util.*;
2
3 class Student {
4     int id;
5
6     Student(int id) {
7         this.id = id;
8     }
9
10    @Override
11    public boolean equals(Object obj) {
12        Student s = (Student) obj;
13        return this.id == s.id;
14    }
15    // hashCode() is intentionally NOT overridden
16 }
17
18 public class Main {
19     public static void main(String[] args) {
20
21         Student s1 = new Student(101);
22         Student s2 = new Student(101);
23
24         System.out.println("equals(): " + s1.equals(s2));
25
26         HashSet<Student> set = new HashSet<>();
```

Output

Clear

```
equals(): true
HashSet Size: 2
HashMap Size: 2

=== Code Execution Successful ===
```



Main.java

Main.java

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```
14 // hashCode() is intentionally NOT overridden
15 }
16
17
18 public class Main {
19     public static void main(String[] args) {
20
21         Student s1 = new Student(101);
22         Student s2 = new Student(101);
23
24         System.out.println("equals(): " + s1.equals(s2));
25
26         HashSet<Student> set = new HashSet<>();
27         set.add(s1);
28         set.add(s2);
29         System.out.println("HashSet Size: " + set.size());
30
31         HashMap<Student, String> map = new HashMap<>();
32         map.put(s1, "First");
33         map.put(s2, "Second");
34         System.out.println("HashMap Size: " + map.size());
35     }
36 }
```

Run

Output

Clear

```
equals(): true
HashSet Size: 2
HashMap Size: 2

=== Code Execution Successful ===
```